BIOLOGICTRAJ · PIPELINE AND SYNTHETIC BENCHMARK · ALL DATA SIMULATED

Testing the pipeline before the patient data arrive

PRISM-like PBMCs with planted answers: 8 patients, 3 visits, ~89,000 cells.

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The pipeline

1 · CHECK — **Stress screen**
Heat-shock vs immediate-early genes

2 · PSEUDOBULK — **Summed counts**
Per sample and cell type

3 · COUNT MODEL — **edgeR / limma-voom**
Patient blocking; tests each visit step

4 · TOPOLOGY — **LongOmicsTraj**
Shape of change, kept only if significant

5 · SCORE — **Against the truth**
Planted shapes known

SCENARIO A · DRUG
NF-κB/AP-1 genes fall at 6 months as a **real drug effect**.

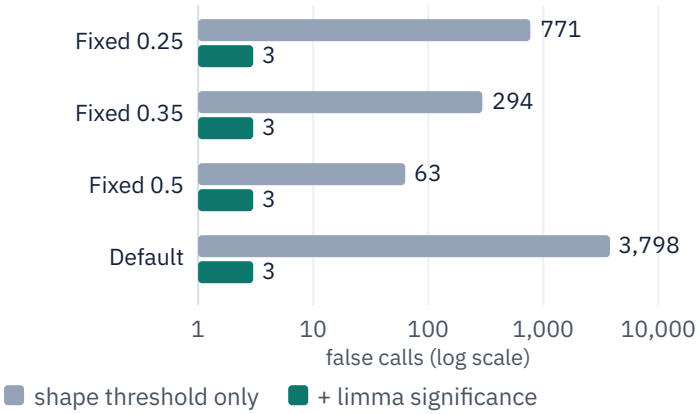
SCENARIO B · STRESS BATCH
Same genes fall, but only because 6-month samples were **handled more gently**.

553 / 553
planted trajectory shapes recovered (scenario A)

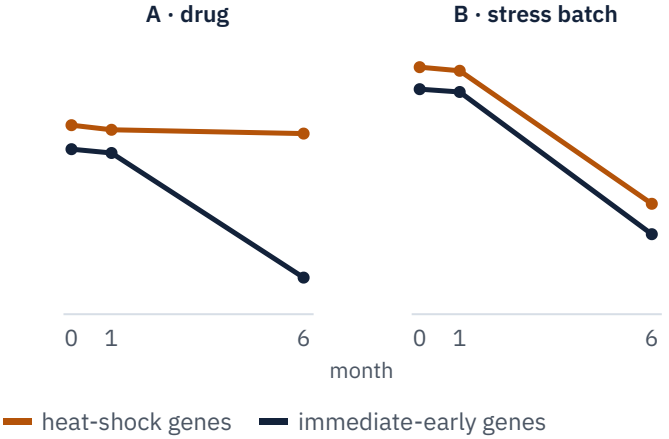
3 vs 294
false calls in 9,582 unchanged genes, with vs without significance filter

14 / 14
cell types flagged by the stress screen in scenario B; none in A

Significance matters more than the threshold



Only the stress screen spots the artefact



What we learned

1 · DEMAND SIGNIFICANCE
Keeping a step only when limma supports it removes nearly all false shapes, at any threshold.

2 · STATISTICS CAN BE FOOLED
465 of 468 artefactual stress-gene trajectories pass the filter. Heat-shock genes give it away.

3 · DEPLETED CELLS VANISH
Anti-IL-5 depletes basophils too far to measure their genes; composition analysis must cover them.

An upper bound: no ambient RNA, doublets or batch noise; true cell labels. **Next:** annotation, composition, scGPT and PerturbGen, then PRISM.

Design after Park et al., *Exp Mol Med* 2024 (PRISM). Tools: edgeR, limma, LongOmicsTraj (bioRxiv 2026). Stress genes after van den Brink et al., *Nat Methods* 2017.

CODE
github.com/nzkermani/BiologicTraj